Inputs

Stock concentration	Unit
Target concentration	Unit
Final solution volume	Unit

Use a decimal point in English. Scientific notation, such as 1e-3, is supported.

Example inputs. Replace these values with your own laboratory quantities.

Calculated results

Stock volume required	0.4 mL
Final solution volume	2 mL
Ideal make-up volume	1.6 mL

Ideal mass balance, no material loss. Use final solution volume. Values based on nominal mass do not establish measured content, purity, solubility or handling conditions.

Displayed values are rounded to at most eight significant digits. They do not establish measurement precision.

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▼ Formula and assumptions

$c_1 \times V_1 = c_2 \times V_2$

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Laboratory calculation
Calculation mode: Dilution
Formula and assumptions: c₁ × V₁ = c₂ × V₂
Final solution volume: 2 mL
Stock concentration: 5 mg/mL
Target concentration: 1 mg/mL
Stock volume required: 0.4 mL
Stock volume required: 400 µL
Final solution volume: 2 mL
Ideal make-up volume: 1.6 mL
User-supplied quantities
Ideal mass balance, no material loss. Use final solution volume. Values based on nominal mass do
not establish measured content, purity, solubility or handling conditions.
Displayed values are rounded to at most eight significant digits. They do not establish
measurement precision.
Formula version: laboratory-v2
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Calculate mass concentration, dilution and aliquots. Results describe ideal mass balance, not measured material content.